

CNA Driver and Probable Classification Report

Sample: **DDLPS_1b** | Assay: low-pass WGS / SAMURAI CNA codification

1 - PROBABLE CNA CLASSIFICATION

Probable CNA classification	Sarcoma/GIST-context CNA pattern
Probable CNA score	44
Probability estimate	0.286
Rule-based CNA class	MDM2_CDK4_gain_amp_CNA_pattern
CNA burden class	CNA-low
N CNA events	9
Altered genome size (Mb)	25.64

Plain-language interpretation for clinician review

Clinician-readable interpretation	Sarcoma/GIST-context CNA pattern
Main copy-number evidence	12q15 MDM2/CDK4-region gain or amplification
Why this pattern was assigned	--sample_set sarcoma was supplied, so interpretation was restricted to that context. Compatible CNA features detected: 12q15_MDM2_CDK4. Knowledge-enrichment label: sarcoma context CNA pattern. Literature support contributed 1 points because selected context-relevant references were found for detected CNA drivers; this supports interpretability but is not clinical validation.
CNA burden context	9 high-confidence CNA events; 25.64 Mb altered; burden class: CNA-low.
How to use this result	Use as supportive genomic context for the pathology diagnosis. CNA patterns can prioritize confirmatory tests such as IHC, FISH, karyotype, SNV/SV assays, expression, or methylation profiling when relevant.
Score meaning	Score 44 (probability-like estimate 0.286): The numeric score summarizes CNA burden, driver-region matches, context-restricted classification rules, and literature support when available. It is not a standalone diagnostic probability.

3 - DRIVER CNA REGIONS TO REVIEW

CNA driver / region	Genes	Observed CNA	Evidence tier	Why it matters	PMIDs
12q15 MDM2/CDK4-region gain or amplification	MDM2/CDK4-region	amplification	supportive-CNA	Adds to amplification-rich and cell-cycle CNA features; interpret with tumor type and focality.	26643872, 31189998, 25799988, 9892106, 24824843, 17002787, 21642372, 20196171

4 - SELECTED INFLUENTIAL LITERATURE

paper_rank	feature_display	influence_score	pmid	year	cited_by_count	title	journal	selection_method
1	12q15 MDM2/CDK4-region gain or amplification	88.627	33799733	2021	70	MDM2 Amplified Sarcomas: A Literature Review.		citation_relevance_plus_optional_huggingface_llm_selection
2	12q15 MDM2/CDK4-region gain or amplification	88.0	28351930	2017	690	Hyperprogressors after Immunotherapy: Analysis of Genomic Alterations Associated with Accelerated Growth Rate.		citation_relevance_plus_optional_huggingface_llm_selection
3	12q15 MDM2/CDK4-region gain or amplification	88.0	26643872	2015	91	Genomic landscape of liposarcoma.		citation_relevance_plus_optional_huggingface_llm_selection
4	12q15 MDM2/CDK4-region gain or amplification	84.543	31189998	2019	85	GLI1-amplifications expand the spectrum of soft tissue neoplasms defined by GLI1 gene fusions.		citation_relevance_plus_optional_huggingface_llm_selection
5	12q15 MDM2/CDK4-region gain or amplification	84.431	21151896	2010	62	Genomic aberrations in lung adenocarcinoma in never smokers.		citation_relevance_plus_optional_huggingface_llm_selection

paper_rank	feature_display	influence_s core	pmid	year	cited_by_co unt	title	journal	selection_m ethod
6	12q15 MDM2/CDK4-region gain or amplification	82.376	30704460	2019	37	Integration of genomic copy number variations and chemotherapy-response biomarkers in pediatric sarcoma.		citation_rele vance_plus_ optional_hug gingface_ilm _selection

5 - LIMITATIONS

This clinician-oriented summary is CNA-only. Low-pass WGS can support genome-wide copy-number screening, CNA burden assessment, and driver-region prioritization, but cannot by itself determine SNVs/indels, balanced translocations, gene fusions, methylation class, expression, clonality, or biallelic inactivation. Integrate with histology, IHC, flow cytometry, FISH/karyotype, SNV/SV data, and clinical context.